## Department of Mechanical Engineering

### B.Tech Project Presentation on

# VOLLEY: Design and Computational Feasibility Assessment of a Commanded Electromagnetic CubeSat Deployment Mechanism




Presented By

**Adityavardhan Mishra · Pratham Chawla**

PRN 23070125054 · PRN 23070125029

**Guide: Vikas Gulia**

**Review candidate: 21 October 2026**

## SYMBIOSIS INSTITUTE OF TECHNOLOGY PUNE

# Presentation route

OUTLINE

**01 Introduction**

02 Literature review

03 Research gap

04 Objectives

05 Research methodology

06 Timeline

07 Work progress

**08 Results**

**09 Conclusion**

10 Future work

11 References

# Which missions might value orbital choice?

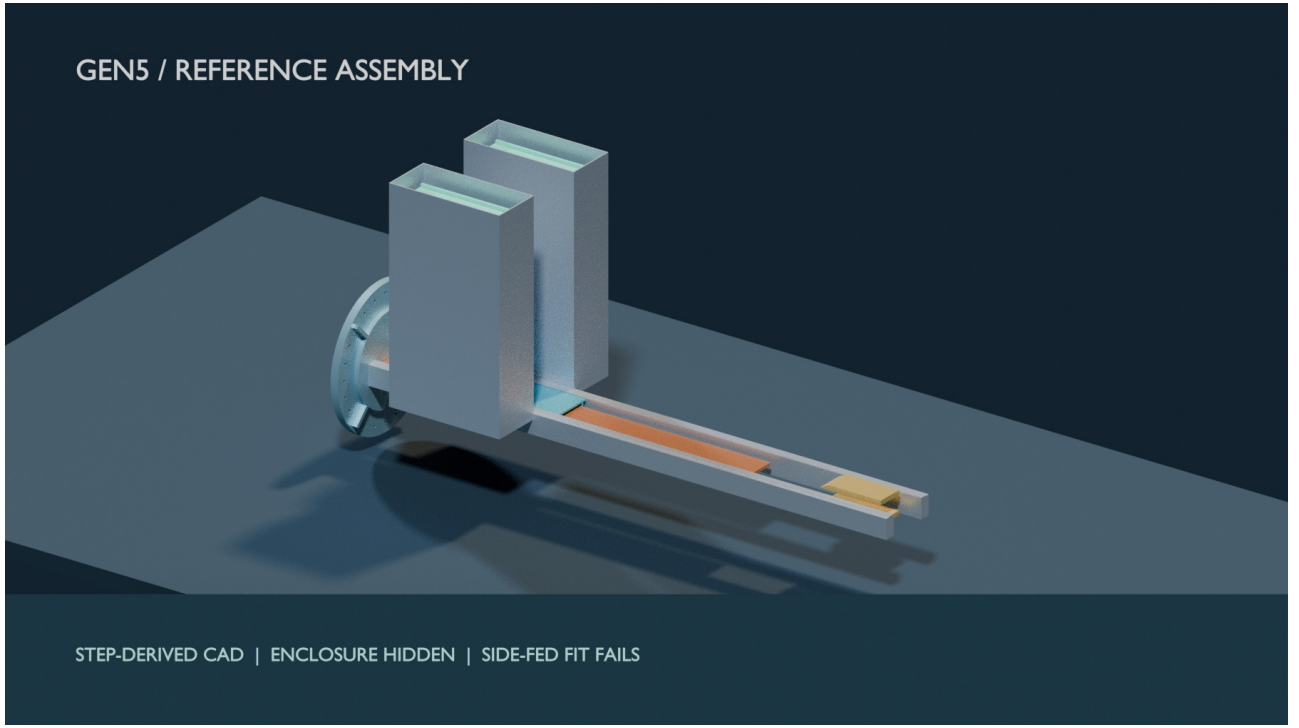
INTRODUCTION

## Repeat fleets

Configurable satellite fleets exist; demand for VOLLEY is unconfirmed.

## Delivery buyer

A carrier or fleet operator might seek several release states from one host.




**STEP-derived Gen5 reference; side-fed placement fails its fit check.**

**Test on one matched mission against springs, host manoeuvres, transport and onboard propulsion.**



NASA 2026 Small Spacecraft SoA; local market audit; STEP-derived Gen5 CAD

# Existing methods solve different parts of the problem

LITERATURE REVIEW

| Approach | Strength | Limit for this thesis |
| --- | --- | --- |
| Spring deployer | Flight heritage; compact; simple | Limited tailored release speed |
| Spring release / drag | Relative state or drag can create phasing | Timing alone, with no relative state change, cannot |
| Orbital transfer vehicle | Larger orbit changes | Greater bus, mass and integration burden |
| Electromagnetic research | Commandable acceleration studied | Armature, high-g or maturity constraints vary |



JAXA; Foster et al. 2018; Feng et al. 2025; Zhao et al. 2022/2025

# A narrower, testable contribution

RESEARCH GAP

**Commanded release for an unmodified CubeSat at moderate acceleration, with the full system counted.**

## Contribution examined

Gen5 reusable-sled linear motor, 3U payload, magazine, pulse power, arrest and orbital utility.

## Boundary

Electromagnetic CubeSat deployment exists in prior work. Payload qualification and physical performance remain unproved.

# The analytical research objectives have outcomes

OBJECTIVES

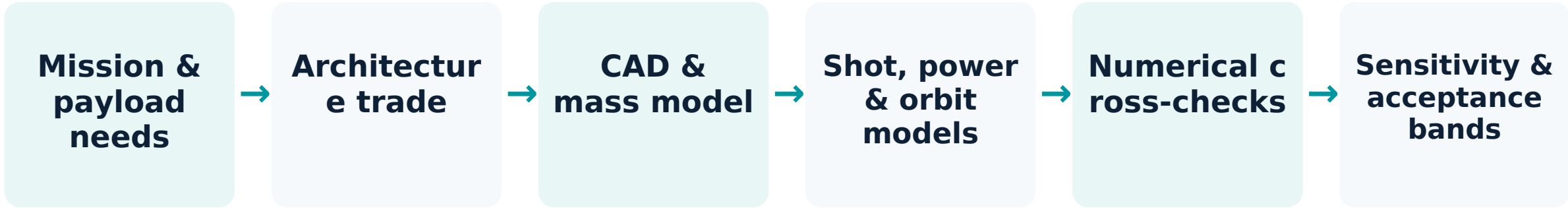
| Research objective | Final thesis outcome |
| --- | --- |
| Design a commandable 3U system | Reference CAD and subsystem concepts documented; feed remains open |
| Calculate shot and control performance | Historical model completed; finite force screen challenges its rating |
| Evaluate orbital utility | Conditional orbit results; real lifetime unvalidated |
| Count full installed burden | Mass rollup completed; 3U parity criterion failed |
| Test credibility of key claims | Numerical cross-checks and limits recorded |



Thesis manuscript, BASELINE.md and PROVENANCE.md; presentation synthesis

# The analysis chain from mission to decision

RESEARCH METHODOLOGY




**Models were challenged and revised; hardware performance remains outside this thesis's evidence.**

# Numerical checks changed the review baseline

RESEARCH METHODOLOGY

| Question | Method / cross-check | Evidence limit |
| --- | --- | --- |
| Field and thrust | Analytic; 2-D FEM; independent 3-D numerical force | Shared ideal inputs; no motor measurement |
| Shot and power | Dynamics; Monte Carlo; ngspice | Assumed components and interfaces |
| Structure and flow | CAD; CalculiX; OpenFOAM | No payload qualification |
| Orbital change | Orbit model; Cowell / GMAT cases | Lifetime depends on atmosphere |

# Project evolution: documented decisions and analysis

TIMELINE

| | | |
|---|---|---|
| 2023 | **Hosted concept** | Free-flyer concept reframed around a host platform |
| Mid-2025* | **LSM direction** | Coilgun study gives way to a linear synchronous motor |
| 2025–26* | **CAD generations** | Geometry and mass models mature through Gen5 |
| Jul–Sep 2026 | **Gen5 evidence** | Cross-checks, defect corrections and thesis record |
| Oct 2026 | **Final review** | Gen5 results, failed criteria and thesis record prepared |

* Approximate historical milestone; not a documented semester deadline.



VOLLEY-thesis/appendix/HISTORY.md; thesis README; PLAN-R2

# 2026 findings changed the review baseline

TIMELINE

| | |
|---|---|
| **Jul** | CAD-derived sled mass replaced an optimistic parametric estimate |
| **Aug** | Depth-resolved thrust calculation lowered the historical model speed |
| **Aug** | Enclosure buildup exposed the full 3U mass penalty |
| **2026** | Independent orbit checks rejected a solar-invariance claim |
| **Final** | Results and failures retained in the submitted thesis record |



HISTORY.md; PROVENANCE.md; thesis manuscript, Limitations

# The current review package is assembled and traceable

WORK PROGRESS




**FreeCAD-linked STEP envelope; internal interference is concealed.**

## Analysis delivered

Shot, circuit, structural, thermal, orbit and alternative-case models.

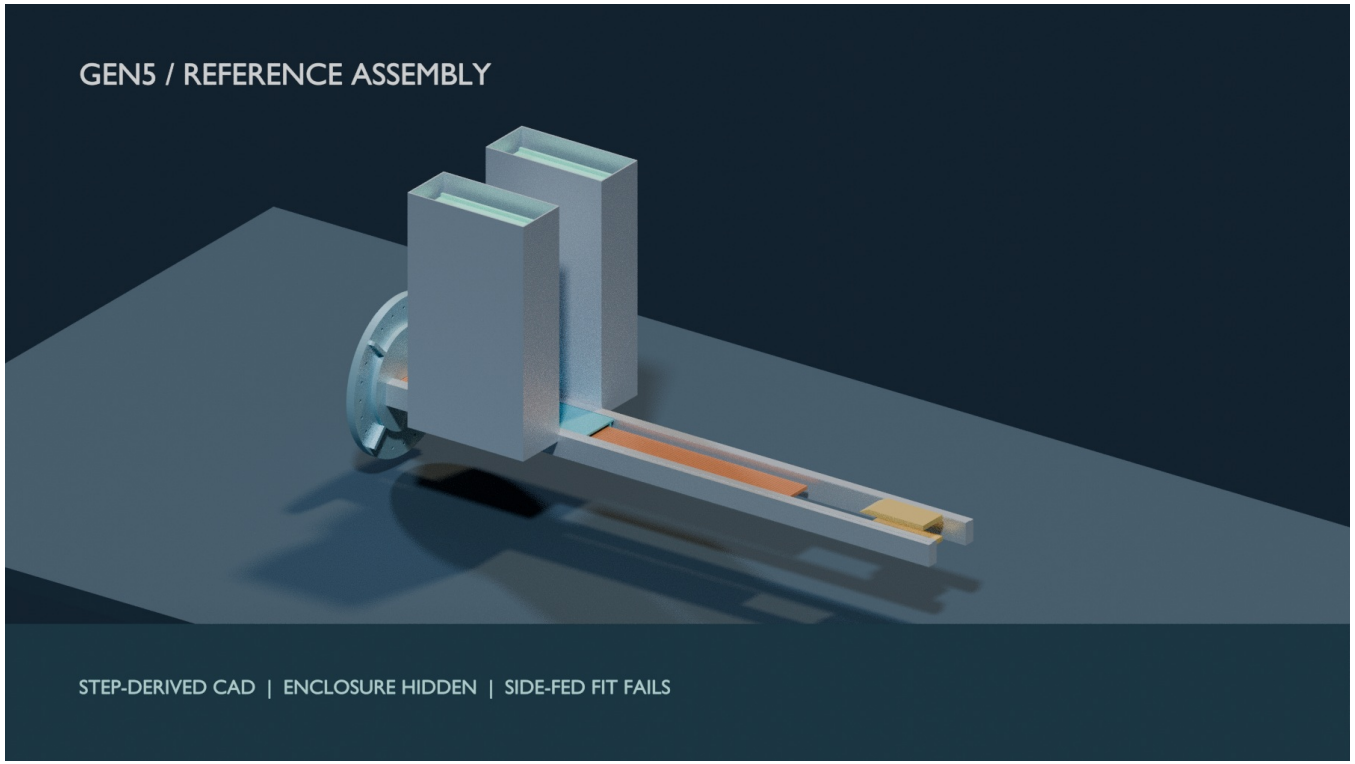
## Evidence delivered

Manuscript, STEP, scripts, numerical checks and defect register.

**Fixed computational study with failed design gates; no physical qualification claim.**

# Gen5: fixed reference geometry, feeder fit still open

**RESULTS**

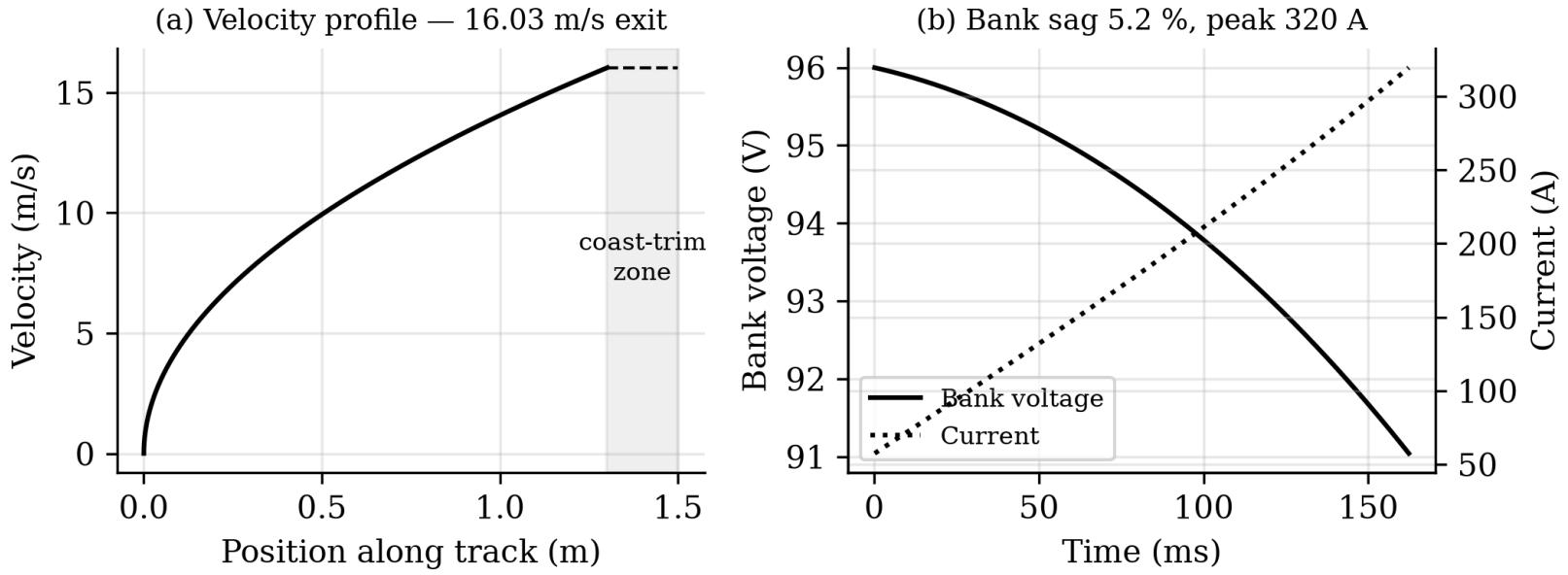



- 1.3 m powered stroke; 1.5 m release station
- Reusable 9.45 kg magnet sled
- Two cassettes; twelve 3U satellites
- FreeCAD STEP assembly: side-fed placement clashes

**Enclosure hidden for visibility; no fabricated article.**

# Historical periodic-model shot: challenged by finite geometry

RESULTS




**16.0 m/s**
exit speed

**10.07 g**
acceleration

**2.782 kJ**
gross draw

**18.8%**
net efficiency

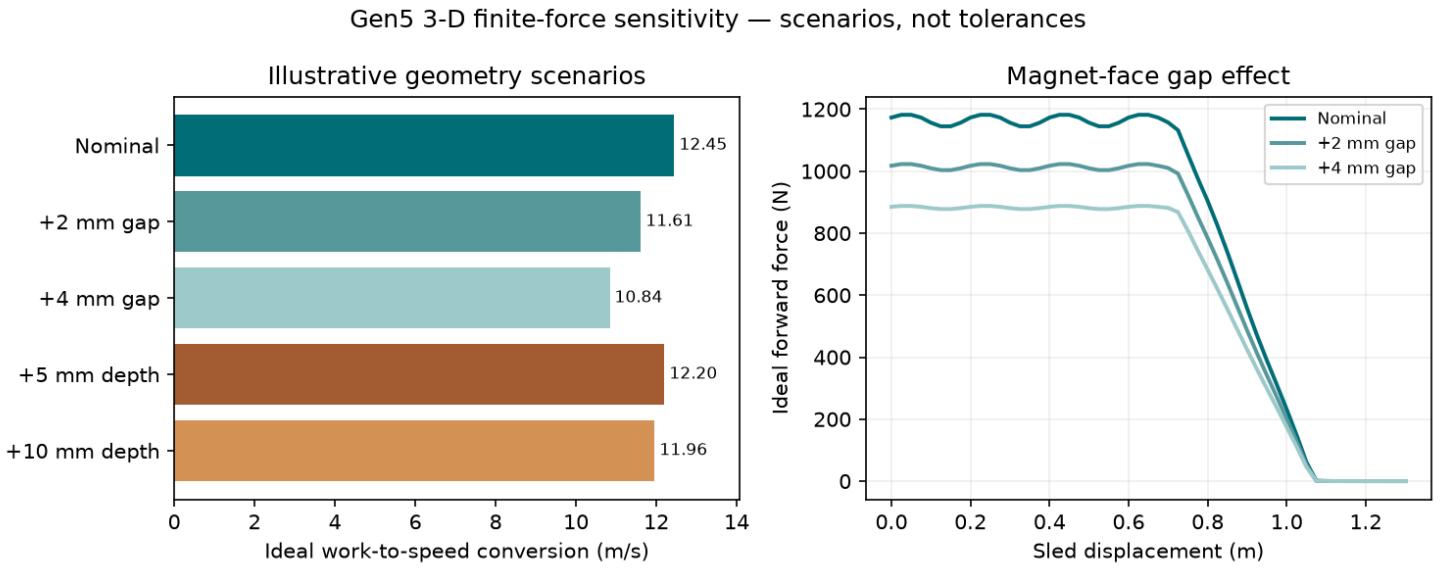
P118 challenges 16.0 m/s; P117 reconciles 124.5 J to assumed model terms.



Thesis BASELINE.md; F01_shot.png; P117 energy audit

# Independent 3-D check confirms ideal finite-force work

**RESULTS**




## 1.042 kJ

Ideal 3-D work: independent surface-charge formulation agrees with P118 under shared inputs.

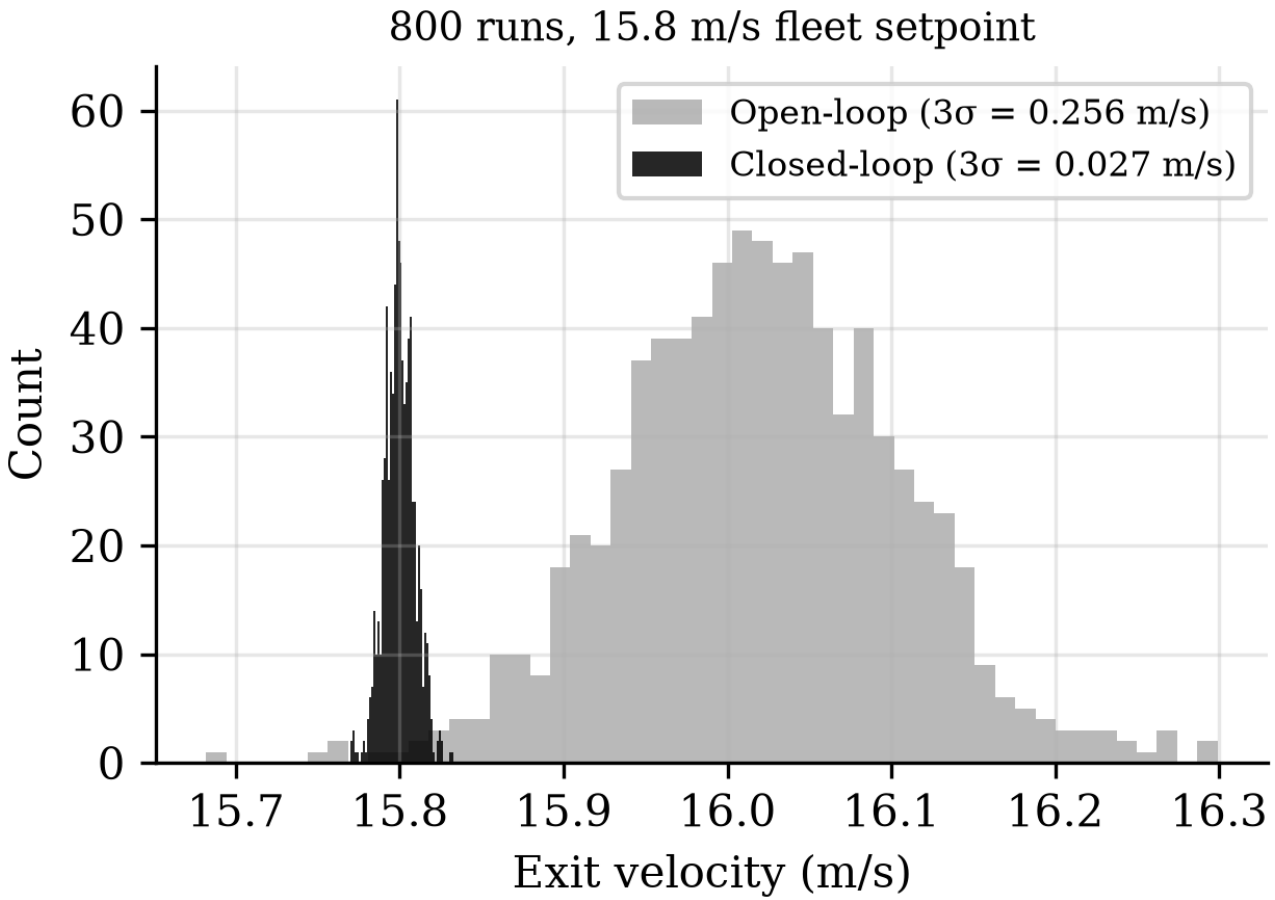
## −13% work

Illustrative 12-to-14 mm magnet gap change; not a measured tolerance or motor rating.

**No measured force, selected inverter or demonstrated release speed exists.**

# Modelled command precision is narrow; it is unmeasured

RESULTS




## 0.0274 m/s

3σ exit-velocity dispersion at a 15.8 m/s fleet setpoint.

## Evidence limit

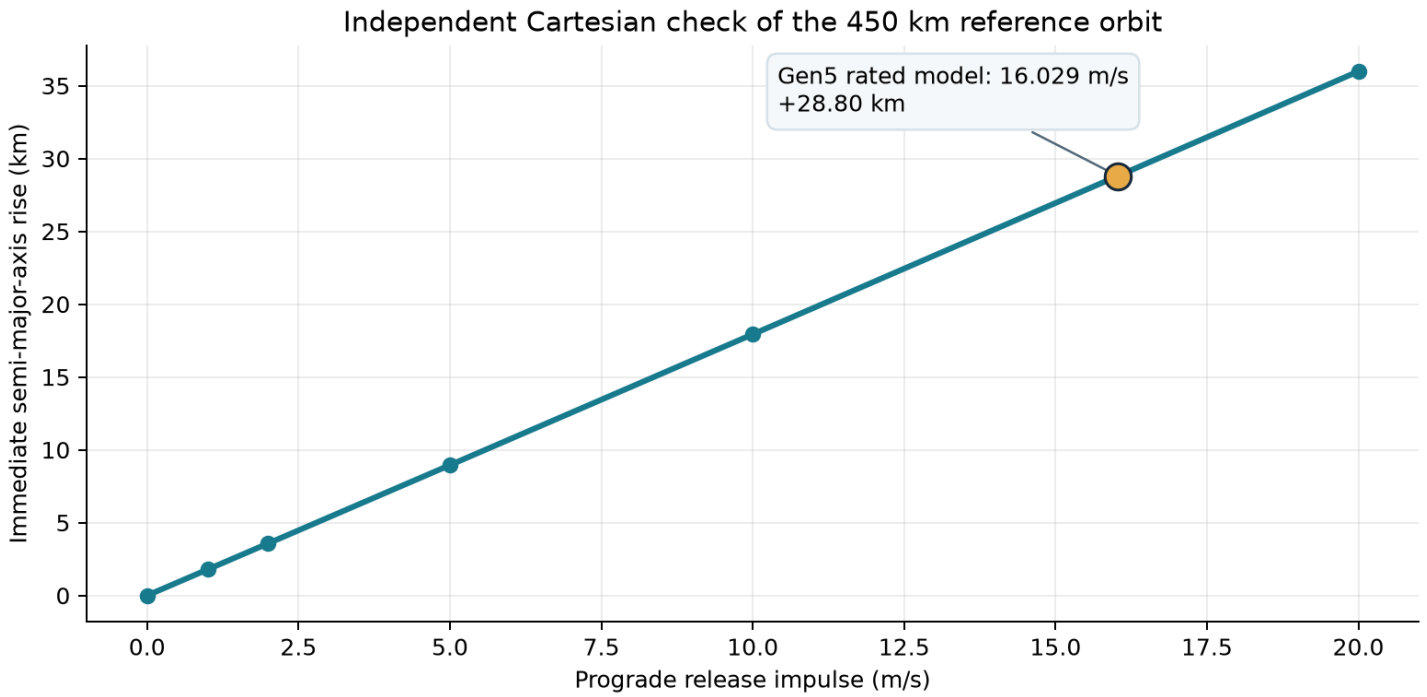
Monte Carlo under modelled sensing and plant tolerances; no measured dispersion.



Thesis BASELINE.md; PROVENANCE.md; F03_mc.png

# Historical orbit input has a separate geometry check

RESULTS




## +28.8008 km

Immediate two-body axis rise for an assumed historical 16.029 m/s input.

## 1.602 model

P134 checks the shared static atmosphere; actual lifetime and achieved release remain unvalidated.

**Persistent phasing requires a relative state change; this result concerns orbital energy.**

# Gen5 fails its 3U mass comparison

RESULTS

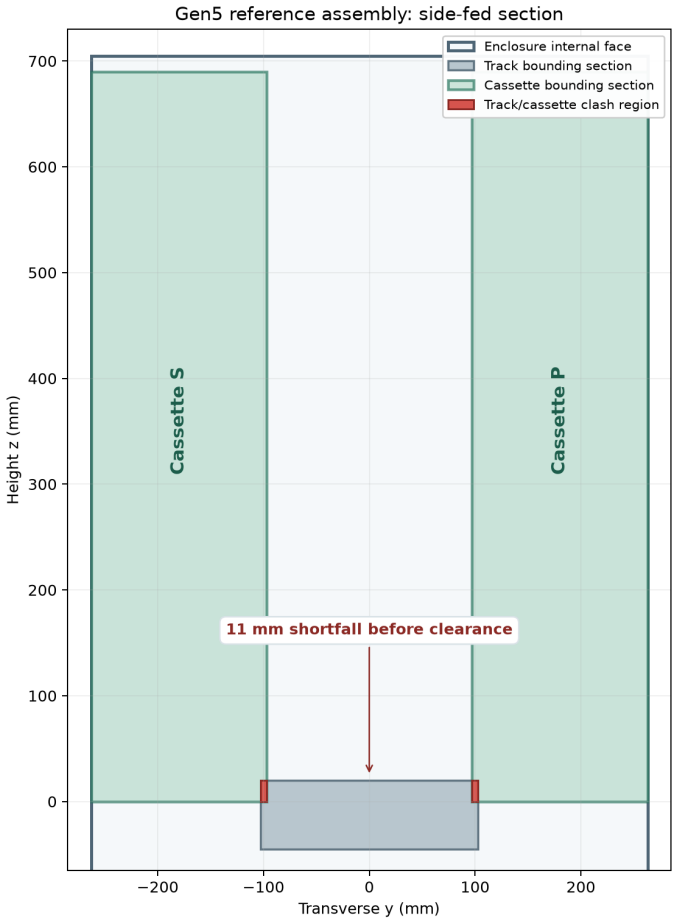
**Modeled dry mass, 12 slots** — **126.6 kg**




**1.76× heavier per 3U payload; the preset 15% parity band fails.**

# The current side-fed assembly fails a geometric fit check

**RESULTS**




## 11 mm short

537 mm of track and cassette width inside 526 mm of usable enclosure width.

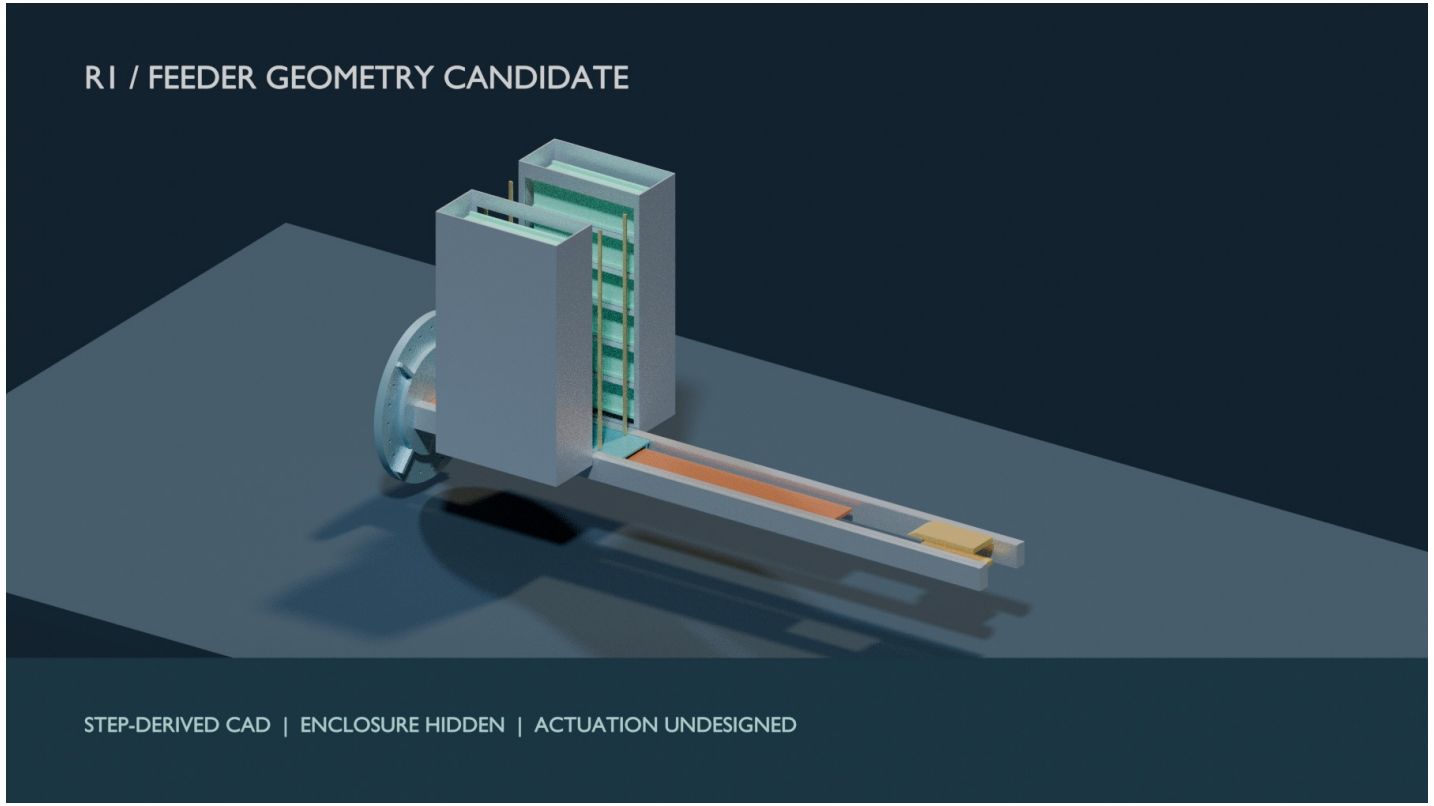
## 32,915 mm³

Cassette/track clash per side. Full audit: 17 intersecting pairs, including sled/track and sled/stator.

**Reference placement only: a new feeder or enclosure would need a new configuration and rerun.**

# R1 clears cassette width, but still has assembly clashes

RESULTS




## 12 / 12

Selected scripted 3U paths clear; a whole-assembly audit still finds five intersecting solid pairs.

## 570 mm

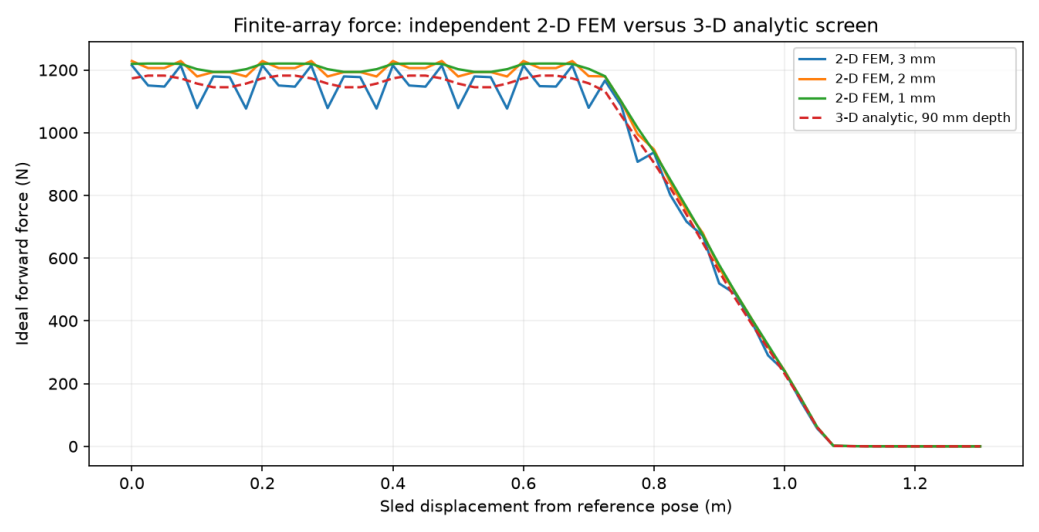
Wider enclosure; 0.5° assumed alignment stack fails its 5 mm gap; lift and retention still open.

**Separate R1 candidate; no change to evaluated Gen5 shot, mass or host-fit claims.**

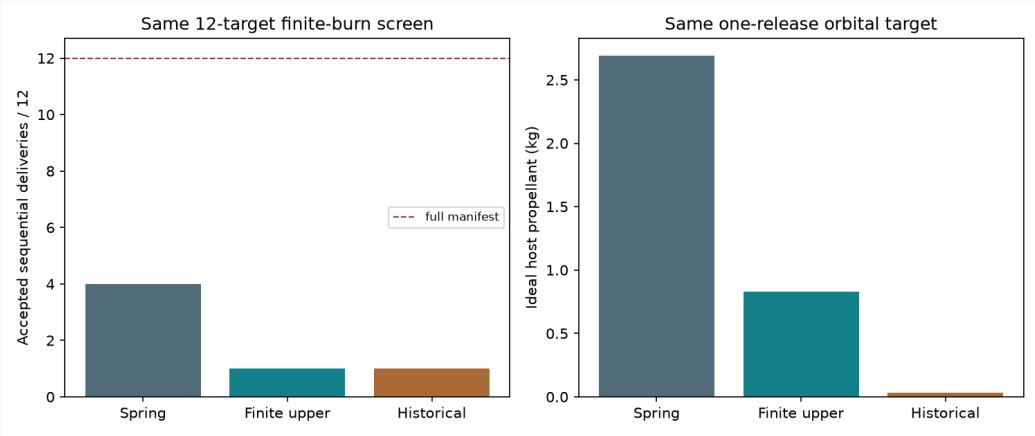
# Computed results: from force to mission decision

RESULTS

## 1. Finite-force geometry screen



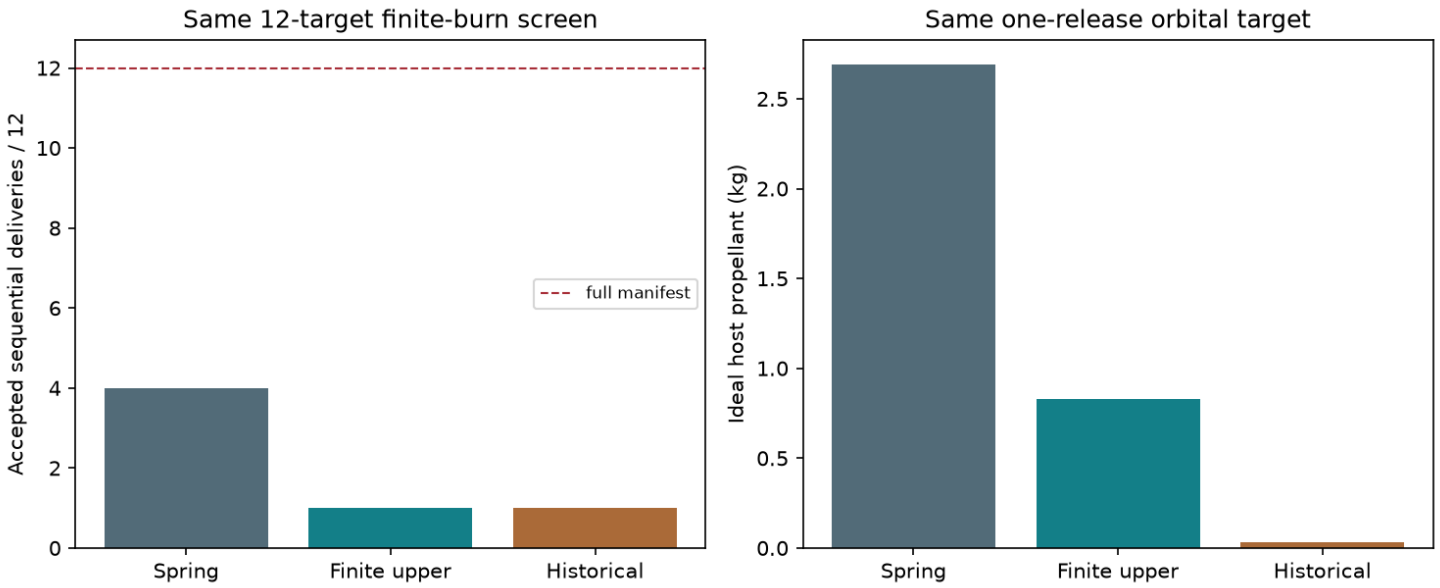

## 2. Matched mission screen




**P125: single-string source power fails at the older commercial ESR bound; no twelve-shot case closes.**

# Matched reference mission finds no twelve-shot closure

**RESULTS**




## +52.7 kg

One-event device plus resized fuel: Gen5 exceeds spring despite an ideal 1.87 kg fuel saving (P123).

## 4 / 1 / 1

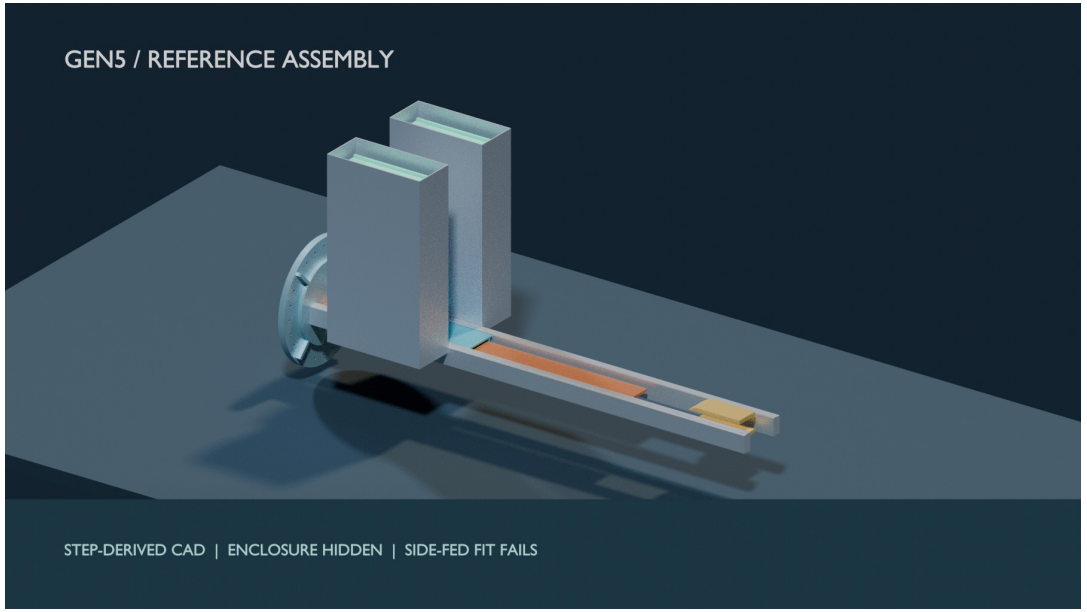
Accepted twelve-shot prefixes: spring, finite Gen5, historical Gen5.

**Assumed host and devices; optimizer result is not an infeasibility proof or product advantage.**

# This Gen5 configuration does not pass selection

CONCLUSION

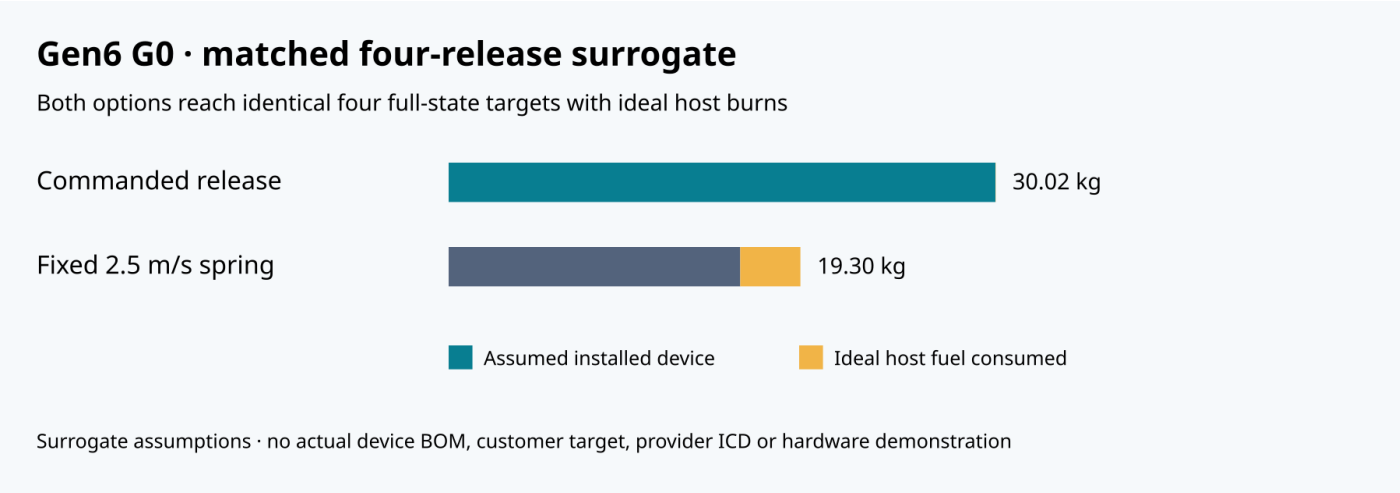
- Historical 16.0 m/s is challenged; ideal 12.448 m/s lacks a sourceable single-string bank.
- The modeled system is 76% heavier per 3U customer and the evaluated side-fed CAD clashes.
- R1 retains sled/track and sled/stator clashes; its assumed 0.5° gap stack fails.
- No matched twelve-shot reference closes; Gen5 is a documented negative design study.




**STEP reference, enclosure hidden; fit fails. No physical test.**

# Gen6 has a testable mission, but no selected article

FUTURE WORK




## Matched mission

Two 1U + two 3U targets: both release choices reach ideal states. Assumed device + fuel favors springs by 10.72 kg.

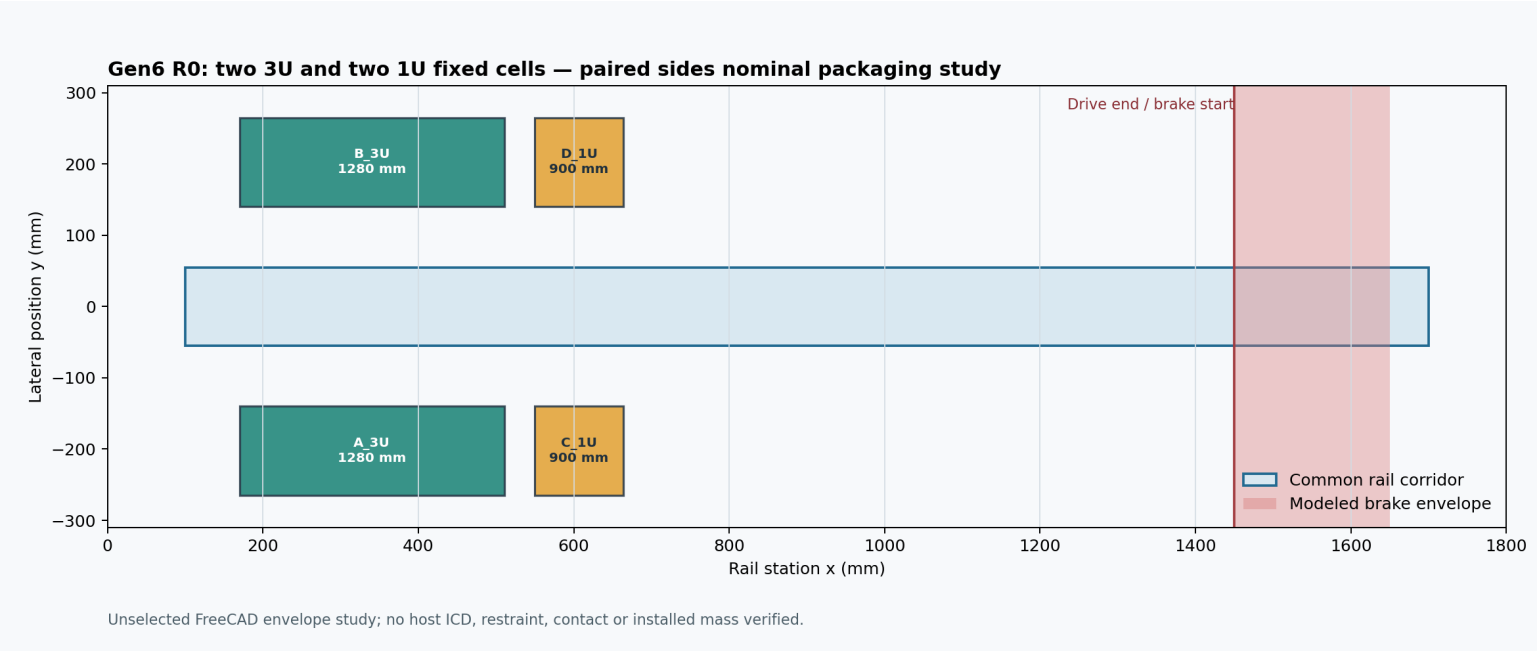
## First-article gap

Two STEP envelopes exist; ~9.35 kg remains for all omitted hardware under a surrogate parity screen.

**1 km/s remains separate long-range research; no command range or motor rating established.**

# Paired cells improve the nominal acceleration path

**FUTURE WORK**




## Staggered

Last cell: 330 mm to brake; 5.56 g ideal mean for 6 m/s.

## Paired

At least 900 mm per cell; 2.04 g largest ideal mean at 6 m/s.

**26 nominal FreeCAD/STEP states pass represented-solid checks; locks, contact, docking and host fit remain open.**

# Next milestone: close G0 and G1 before any build

FUTURE WORK

| | |
|---|---|
| **Specify** | Real customer targets, tolerances and provider interface |
| **Decide** | Complete motor, source, brake, contact and installed BOM |
| **Measure** | Field, contact, release, arrest, loads and cycle behavior |
| **Review** | Drawings, BOM, procedures, uncertainty and review decision |

# Core literature and standards (1/2)

REFERENCES


- [1] JAXA, JEM Payload Accommodation Handbook, Vol. 8: Small Satellite Deployment ICD.
- [2] Cal Poly, CubeSat Design Specification, Rev. 14.1, 2022.
- [3] C. Foster et al., ‘Constellation phasing with differential drag on Planet Labs satellites,’ J. Spacecraft Rockets 55(2), 2018.
- [4] H. Feng et al., ‘Design and reachable domain analysis of on-orbit electromagnetic launcher for CubeSats,’ Int. J. Aerospace Eng., 2025.
- [5] Y. Zhao et al., ‘Design and analysis of a new deployer for the in-orbit release of multiple stacked CubeSats,’ Remote Sensing 14(17), 4205, 2022.

# Methods and project records (2/2)

REFERENCES


• [6] Y. Zhao et al., ‘Simulation analysis and experimental verification of the transport characteristics of a high-volume CubeSat storage device,’ Aerospace 12(6), 466, 2025.

• [7] K. Halbach, ‘Design of permanent multipole magnets with oriented rare earth cobalt material,’ Nucl. Instrum. Methods 169, 1980.

• [8] NASA, General Environmental Verification Standard (GEVS), GSFC-STD-7000A, 2013.

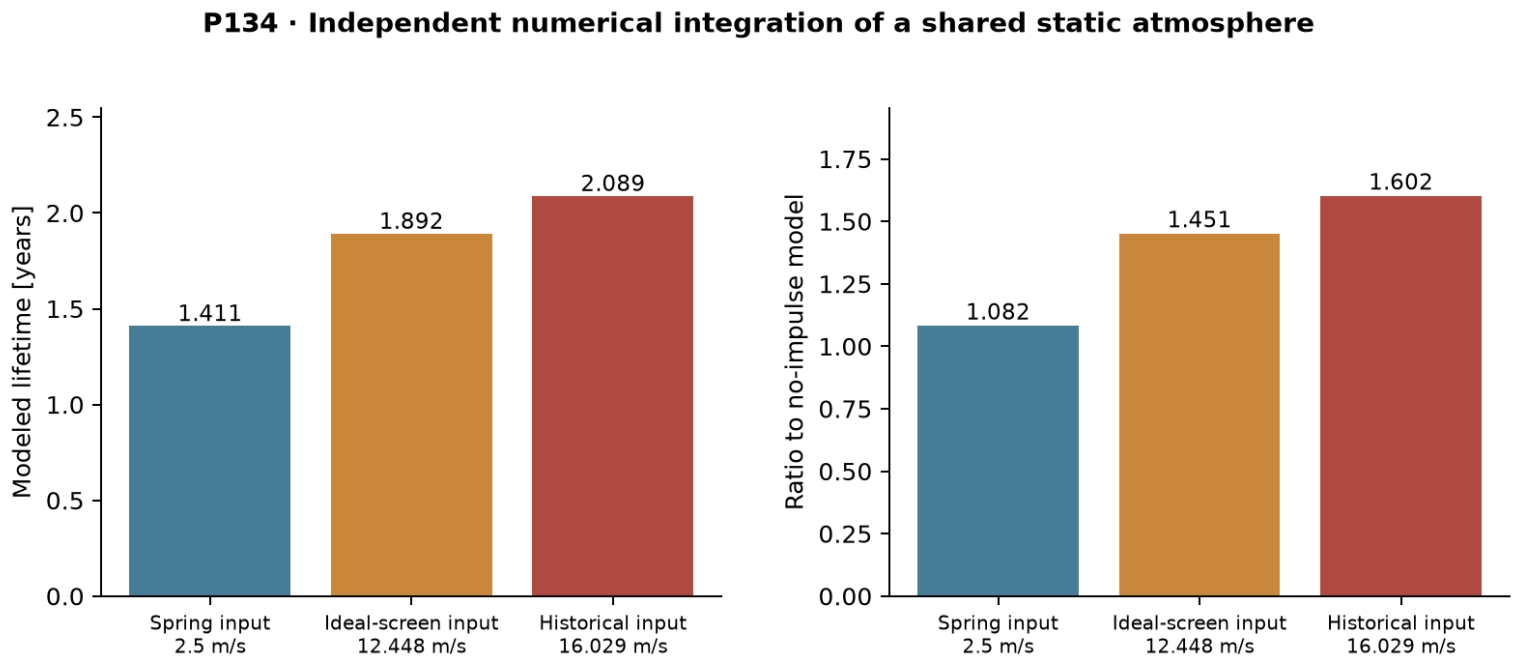
• [9] D. A. Vallado, Fundamentals of Astrodynamics and Applications, 4th ed., 2013.

• [10] A. Mishra, VOLLEY-thesis: manuscript, baseline, provenance and validation records, evidence snapshot 5b92edc, 20 Oct 2026.

• [11] VOLLEY, Gen6 G0 mission and article requirements, 12 Oct 2026 (study requirements).

# What the reconciled numerical record establishes

EVIDENCE APPENDIX




450 km start; BC 61 kg/m²; stop at 120 km perigee. Speeds are imposed inertial inputs.
Numerical agreement does not validate atmospheric lifetime or an achieved release speed.

## 1.602 model ratio

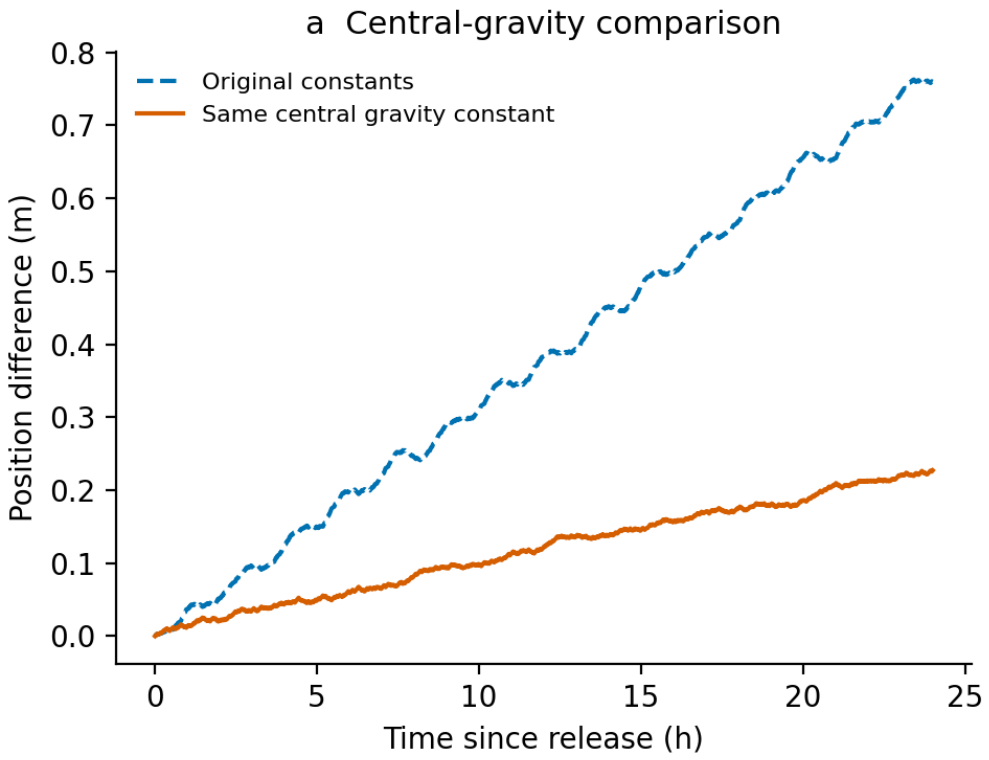
Separate integration, same static atmosphere. Actual lifetime is unvalidated.

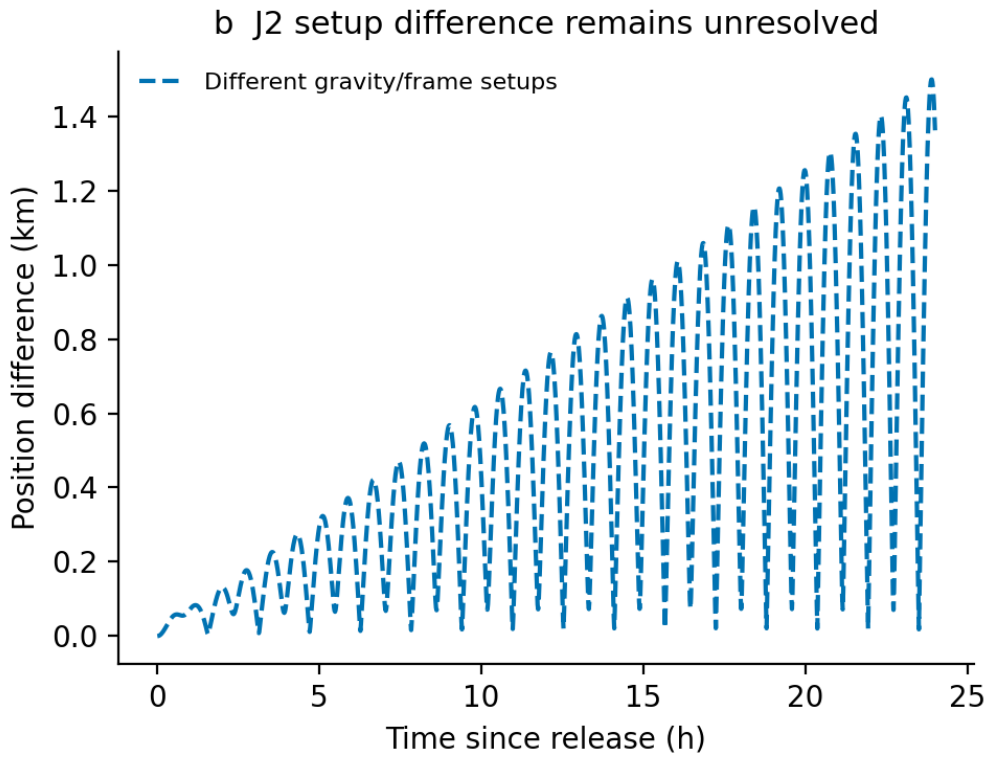
## Failures retained

22 Gen5/R1 intersections; 126.562 kg model; archived mission screens fail. New Gen6 targeting is separately bounded.

# Native solvers: agreement and unresolved differences

**EVIDENCE APPENDIX**



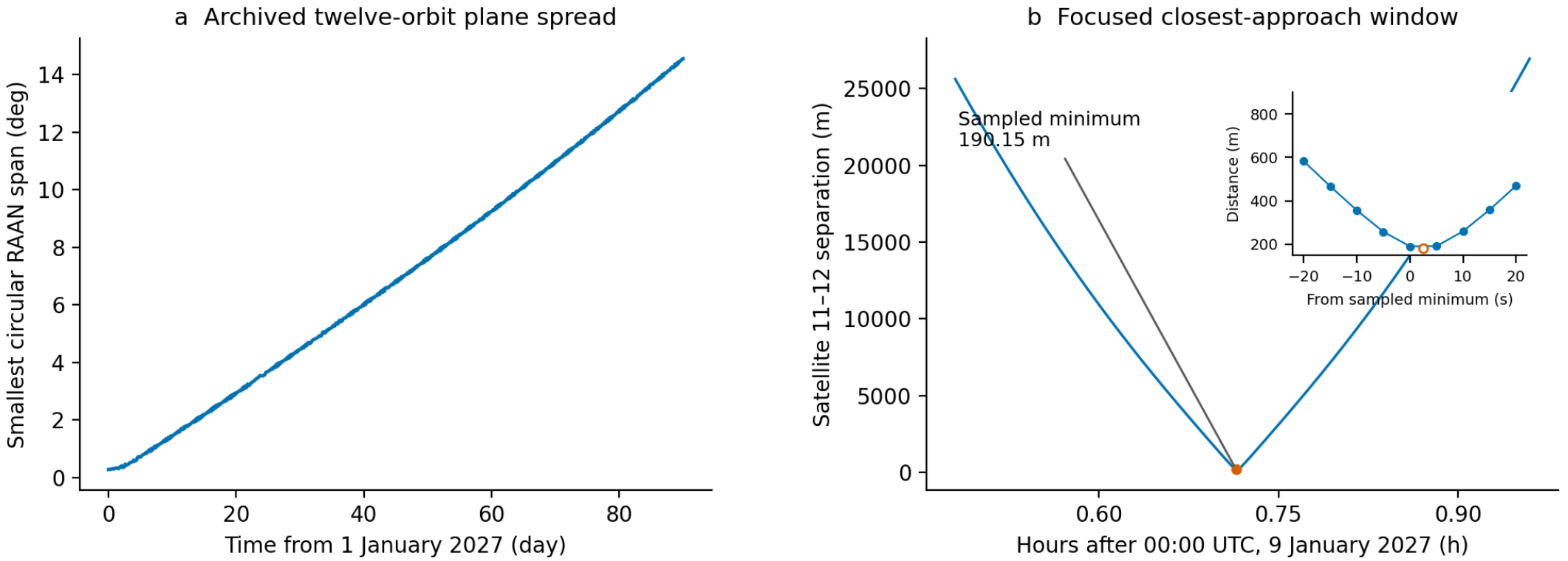




Native GMAT R2026a versus Orekit 13.1.6; identical prescribed initial Cartesian state, 16.029 m/s relative input.
Elapsed output epochs are aligned. The J2 comparison is not a matched-force validation; its residual must not be hidden.

Prescribed states; no release-accuracy or hardware validation. J2 frames are not harmonized.

# Archived campaign: sampled proximity has a limit

EVIDENCE APPENDIX

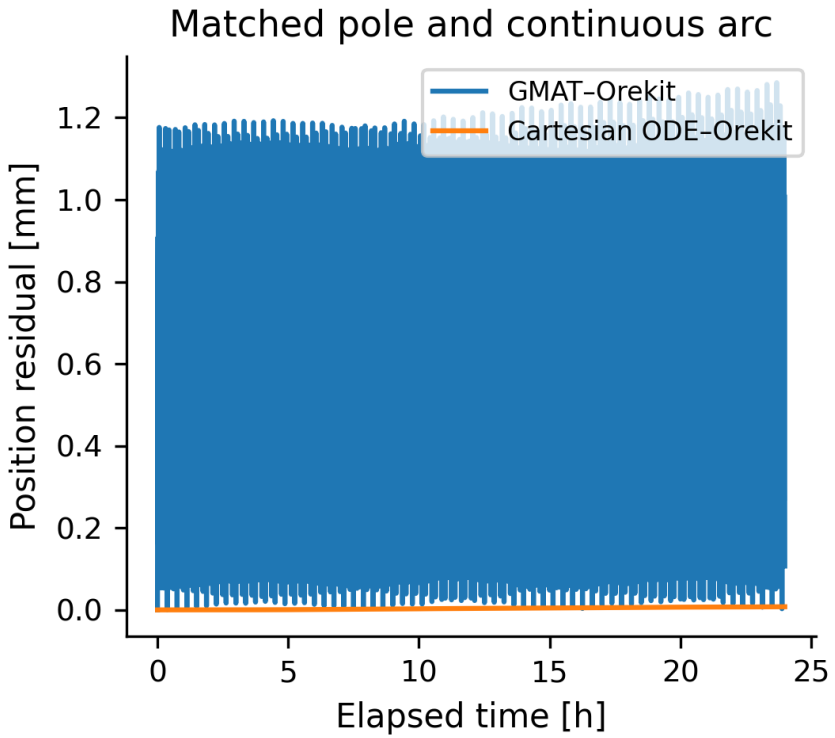


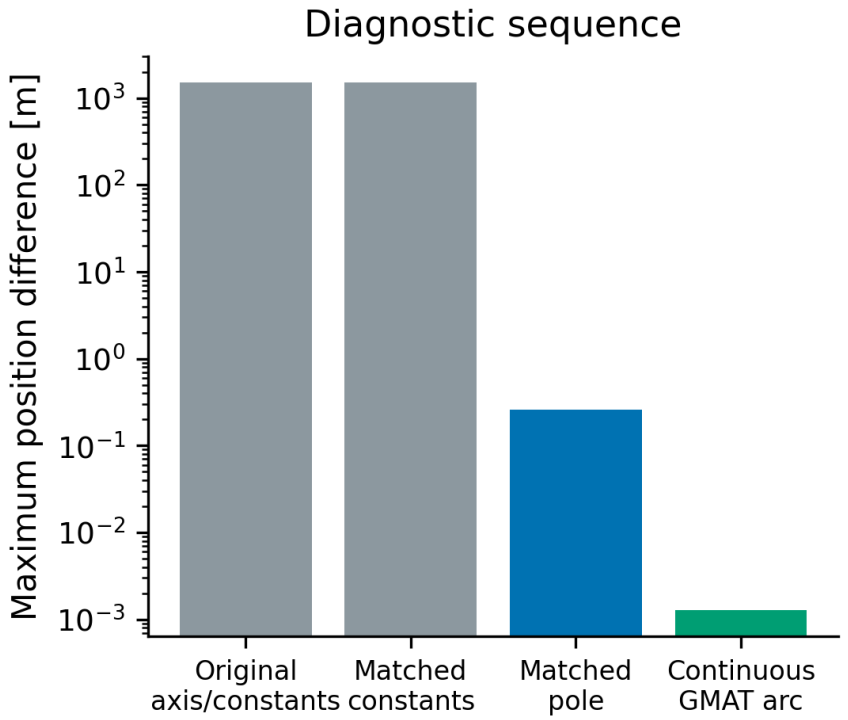

GMAT R2026a archived R1: preloaded post-release orbits, not a coupled host campaign. The left plot is downsampled.
The 190.15 m sampled minimum and 180.29 m linear interpolation do not establish continuous clearance or collision probability.

190.15 m sampled; 180.29 m interpolated. Continuous clearance remains unproven.

# The J2 setup discrepancy now has a bounded resolution

EVIDENCE APPENDIX



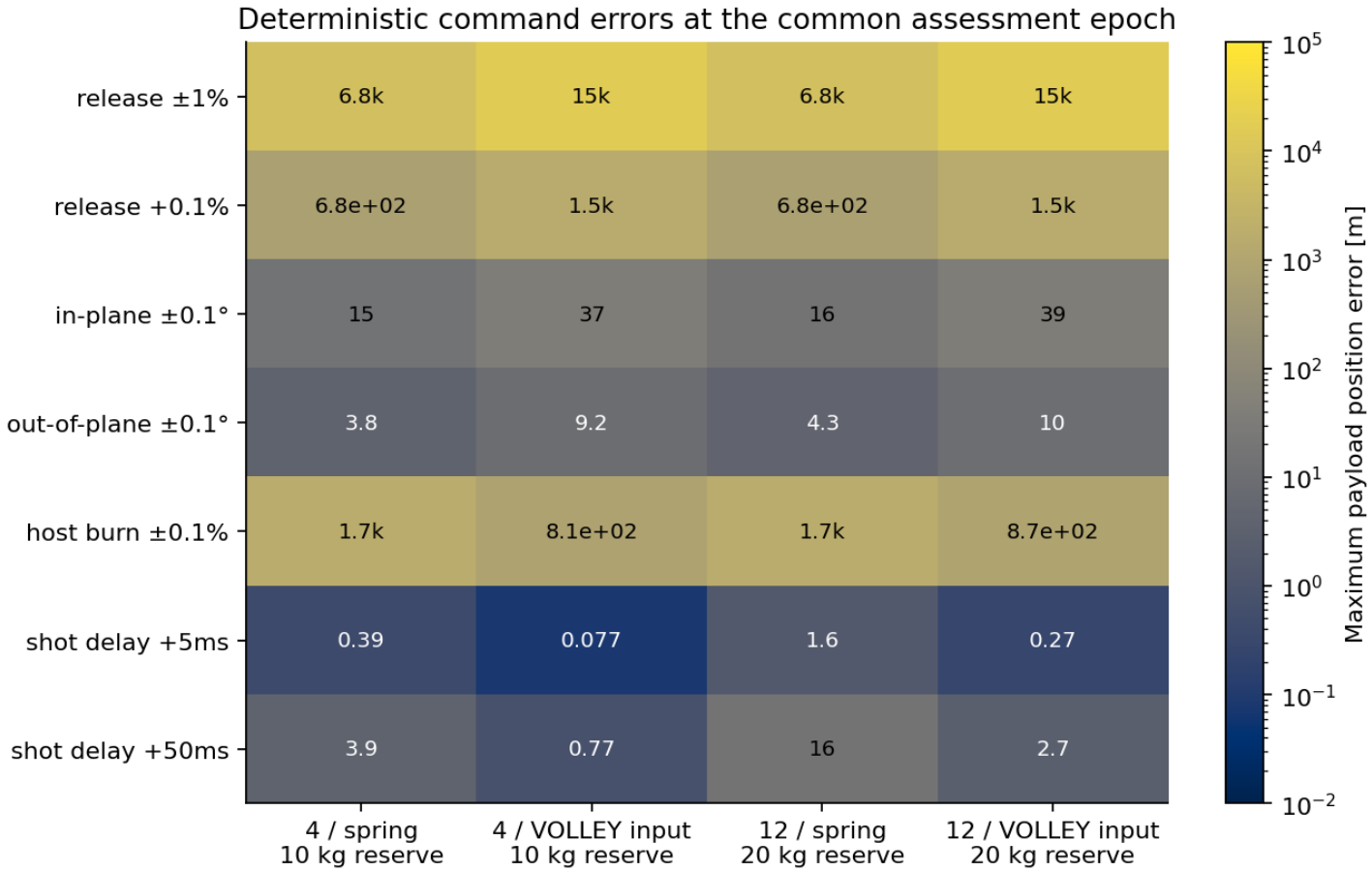




450 km, 51.6°; one day, J2 only. Shared prescribed GMAT EarthFixed pole. First three bars: SciPy versus repeated-stop GMAT; last: Orekit versus continuous GMAT. Different numerical histories are declared; no flight-accuracy or motor validation.

1.285 mm maximum over one day. Shared orientation; no physical orbit or release-accuracy claim.

# Precise delivery needs a measured control budget

**FUTURE-WORK APPENDIX**

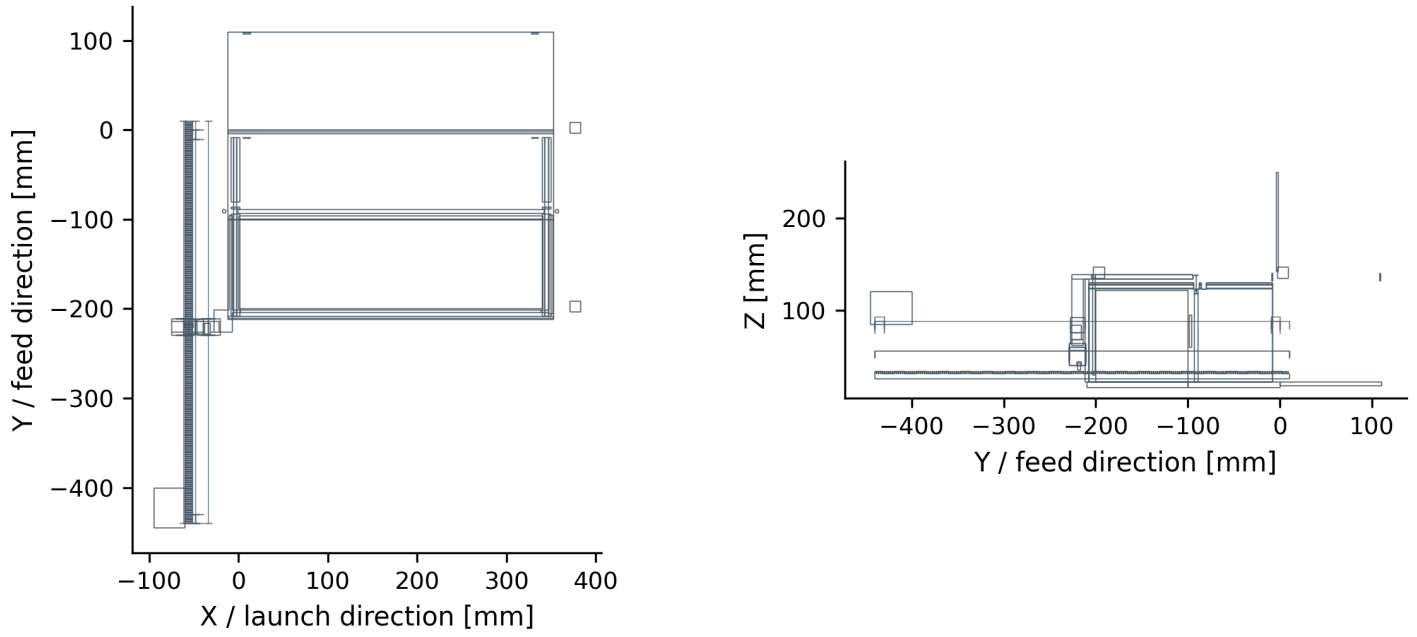



+0.1% imposed release-speed bias gives ~1.53 km error. Numerical agreement is not flight precision.

# Seven feeder concepts: inspectable geometry, open hardware gates

**FUTURE-WORK APPENDIX**

lead screw · 3U nominal stowed geometry



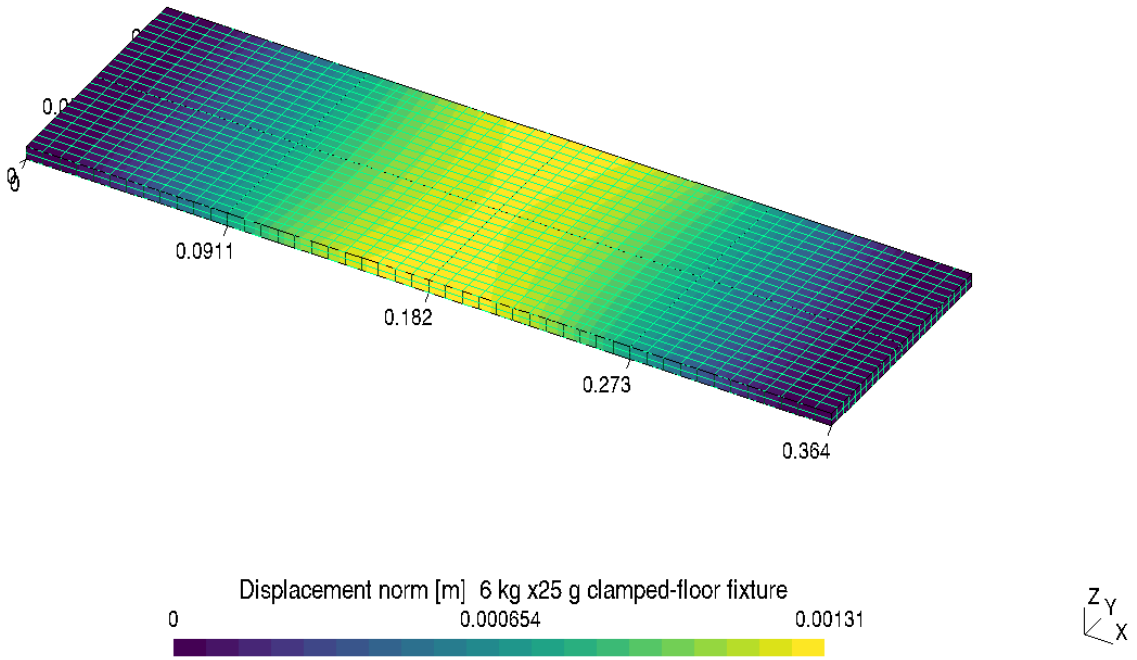

Independent FreeCAD/OCC STEP readback. Orthographic wireframe, not hardware or verified full-cycle motion.
Unselected drive, sensors, contact/load paths and actual host interface remain open.

- 1U/3U CAD and bounded motion checks
- Load, tolerance and fault cases retained
- Arrest-tooth sizing fails adverse cases
- Handoff, latch engagement and installed parts unverified
- No baseline selection or hardware release

STEP-derived orthographic view; no measured reliability or full-cycle qualification.

# Native mesh and result view: keep the fixture explicit

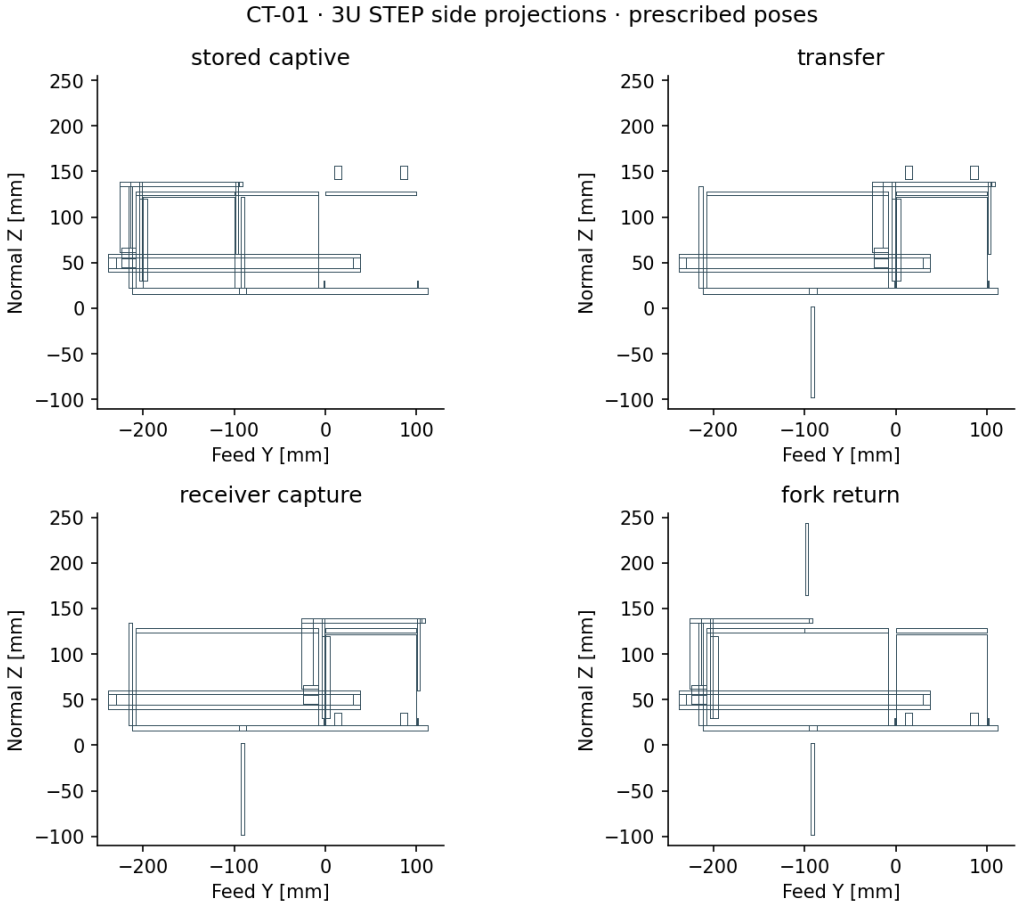
**EVIDENCE APPENDIX**




Displacement norm in metres. Actual joints absent; peak support stress is not converged or a strength pass.

# Continuous feed geometry exposes the remaining safety work

**FUTURE-WORK APPENDIX**

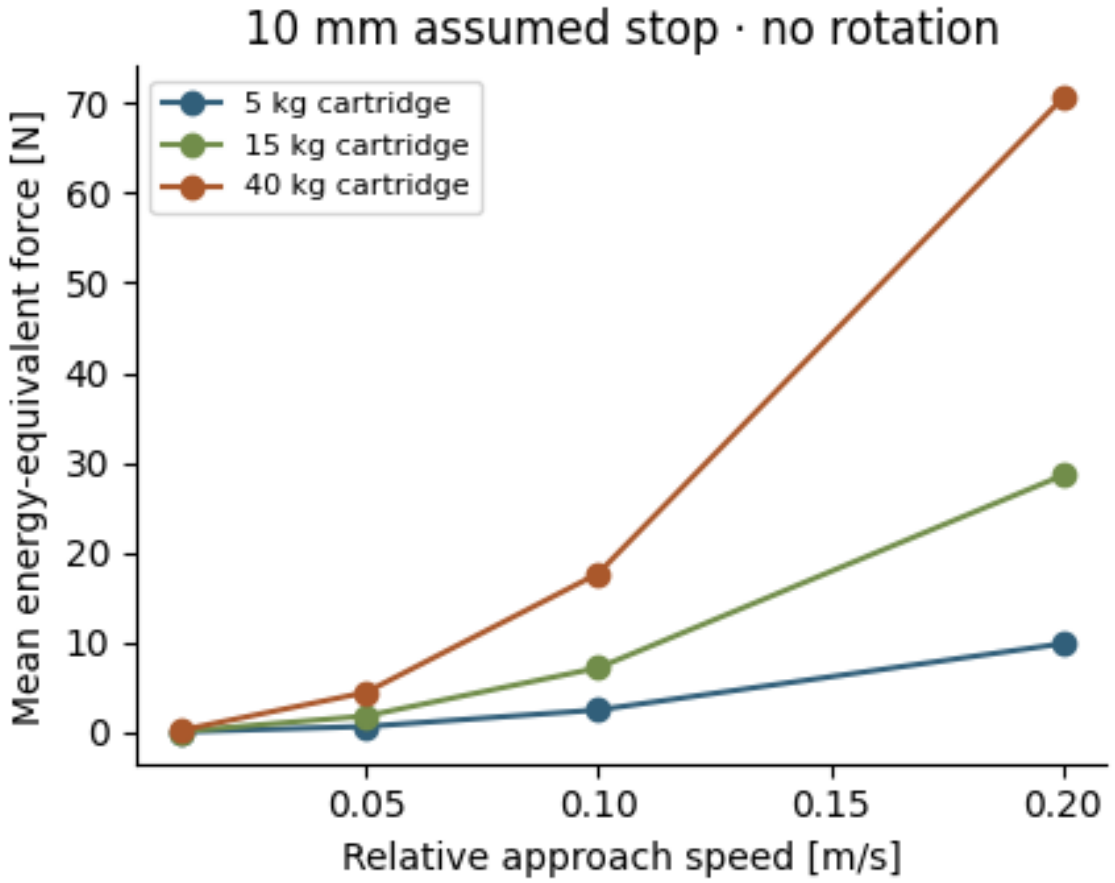


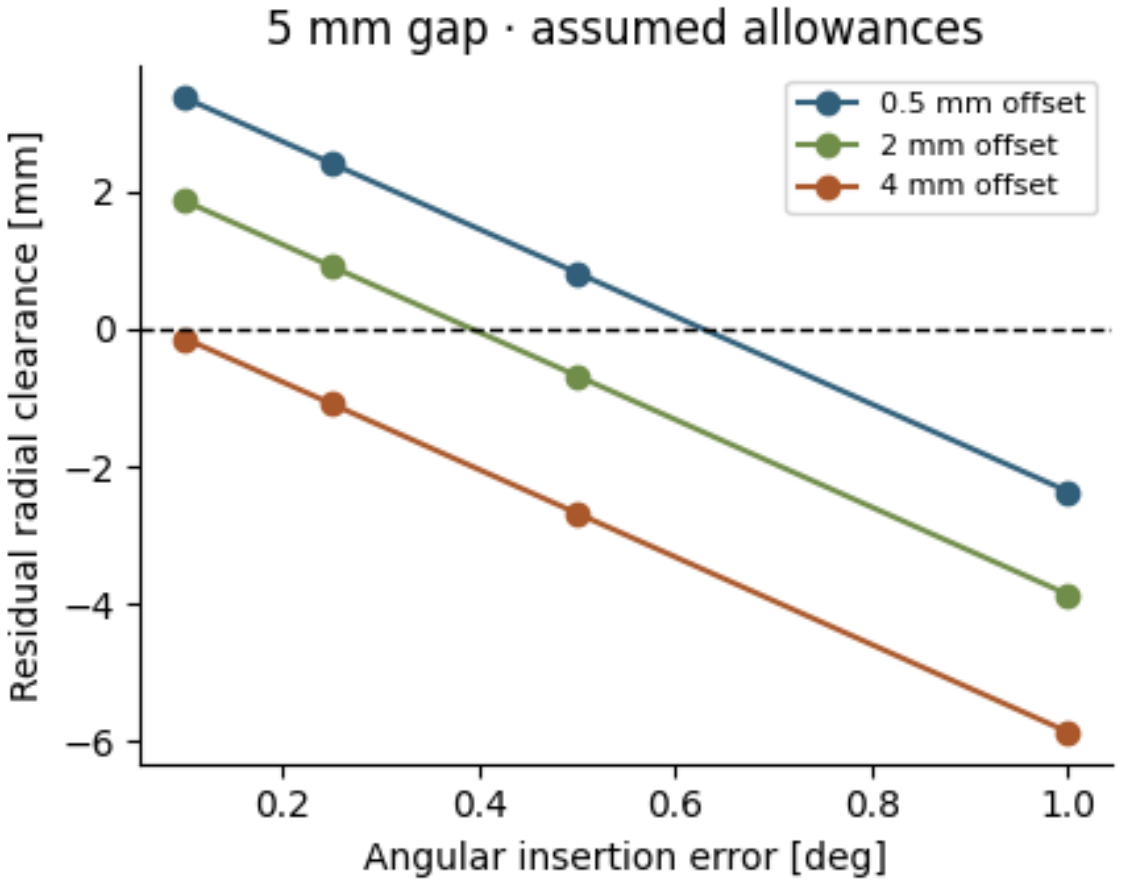

- 20 continuous 1U/3U phase checks
- Downward gate needs under-deck volume
- Stop travel and adverse fit cases fail
- Real latch/custody and drive remain open

Native CAD and nominal translation proof do not establish full mechanism safety or a selected feeder.

# Orbital reload: define the interface before promising a service

**FUTURE-WORK APPENDIX**

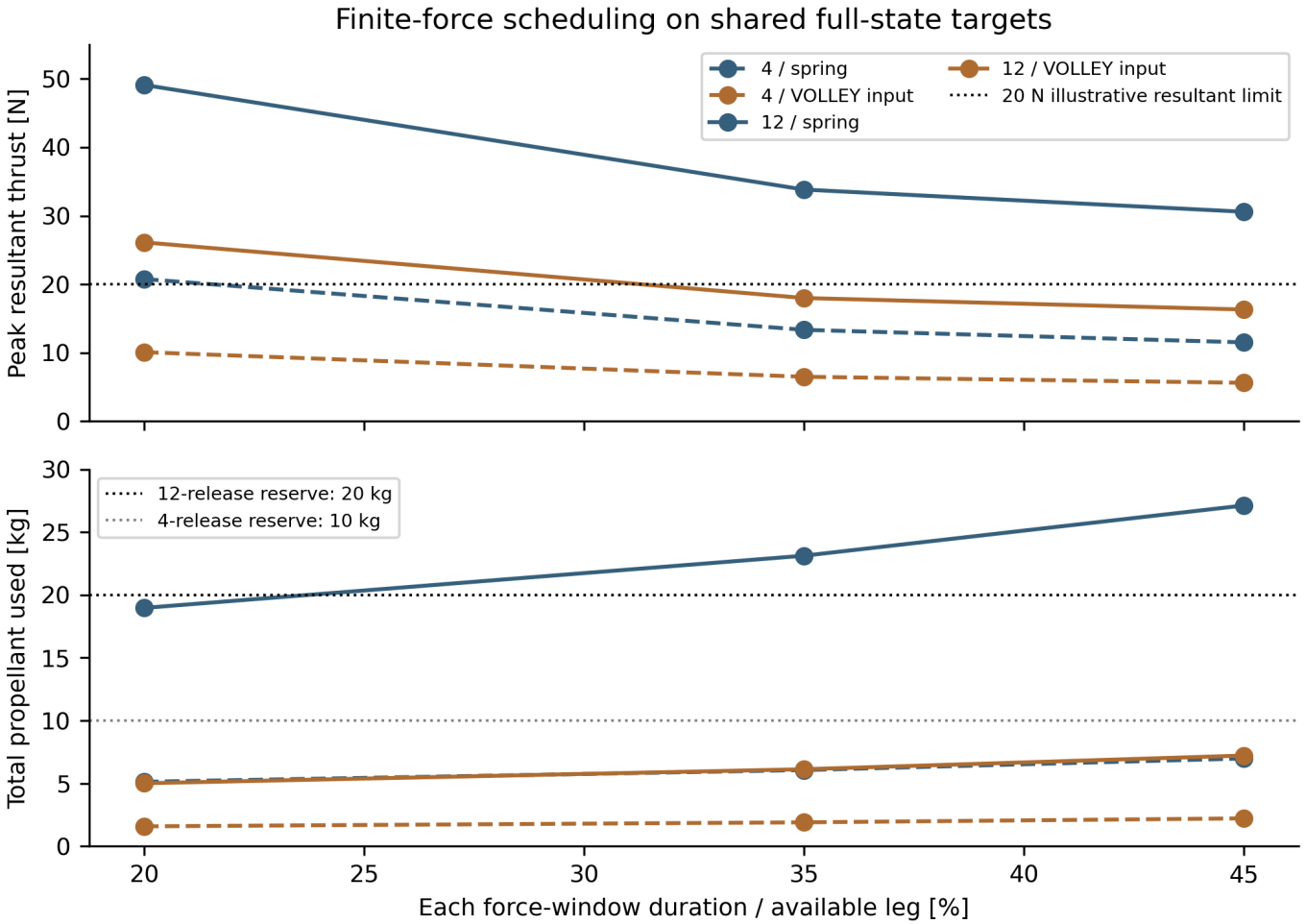



108 capture / 36 fit sensitivities. No integrated reload mechanism; ground loading remains the assumption.

# Finite-duration thrust changes the resource comparison

**FUTURE-WORK APPENDIX**

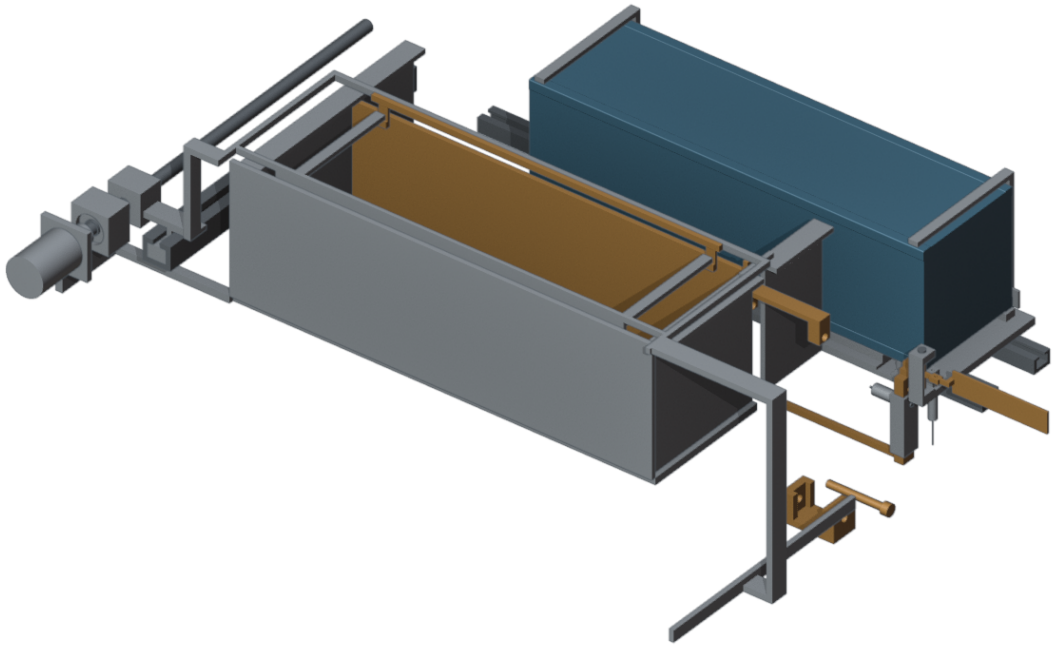



- Four releases fit the 20 N / 10 kg screen
- Twelve spring inputs exceed 20 N / 20 kg
- Both twelve-case policies close with 30 kg / 100 N
- Device mass still exceeds saved propellant
- Nearly 9°/s reversals if one fixed thruster; host allocation unverified

Finite-force translational targets; imposed release speeds. No rated host, motor, flight precision or physical clearance.

# Feeder geometry includes exit release and empty reset

FUTURE-WORK APPENDIX

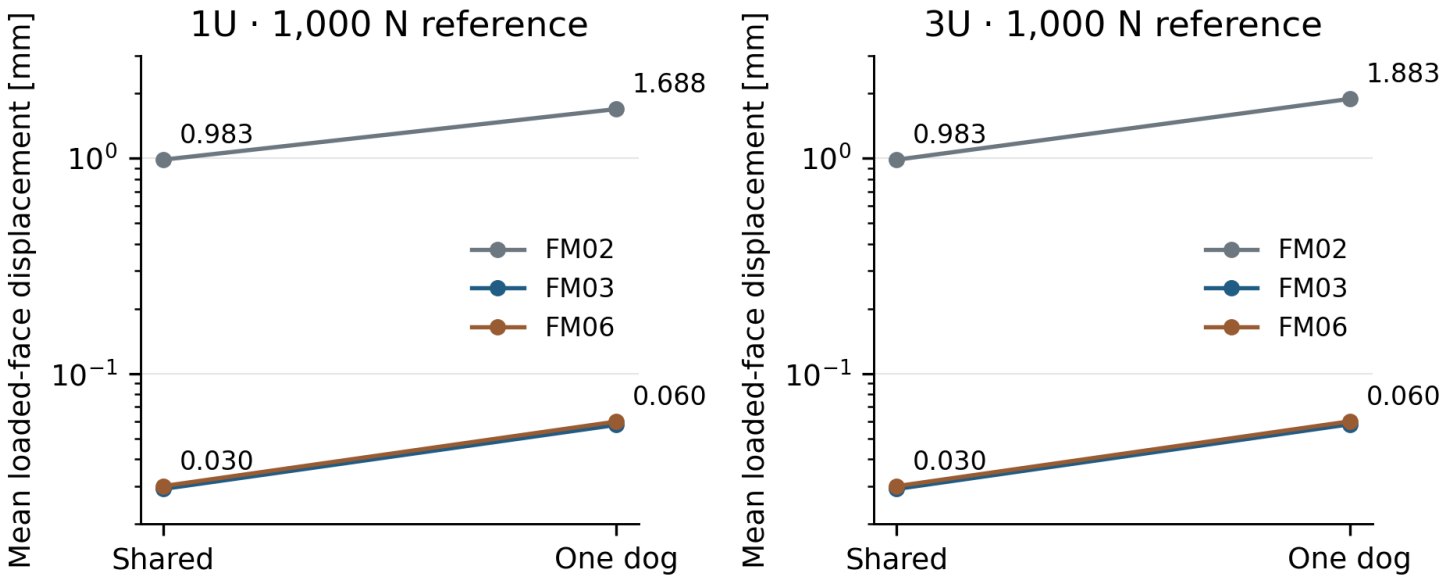


- 46 cycle poses; 34 additional door-angle samples
- 40 continuous prescribed linear sweeps
- 20 assembly STEP files; 74 checked individual parts
- Loaded fits, installed actuation and seven-way downselect remain acceptance gates

Blender Workbench render of actual native CAD. Prescribed geometry, not launch dynamics or hardware qualification.

# Component stiffness improves within an idealized fixture

FUTURE-WORK APPENDIX

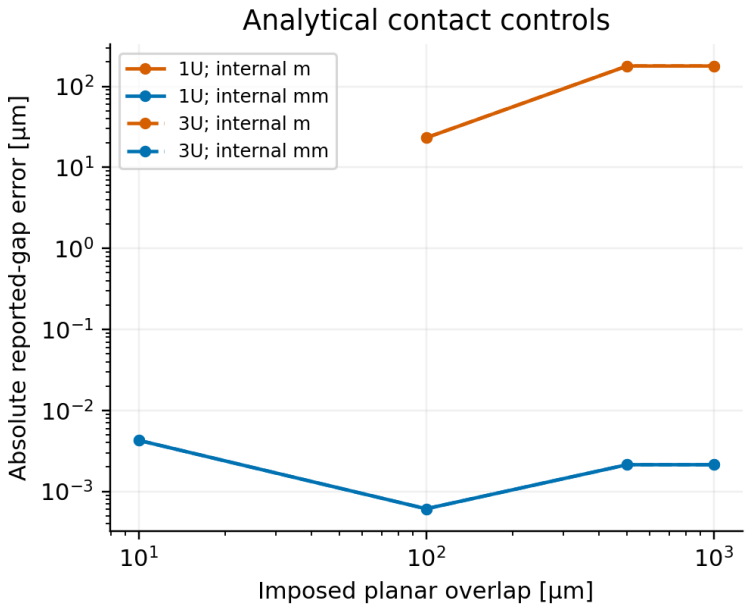


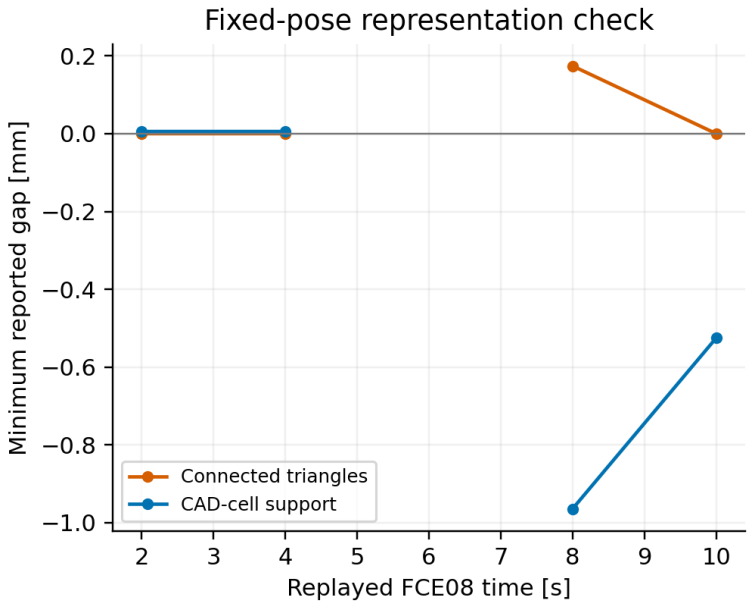

- 12 actual-FM06 plus 40 historical receiver analyses
- 3U one-dog mean: 1.883 to 0.0600 mm
- Clamped bores do not bound assembled stiffness
- 1,117/1,215 adverse guide-fit residuals nonpositive

Assumed elastic properties; no selected material allowable, real joint-contact acceptance or hardware testing.

# Contact representation: calibrated controls, sensitive final pose

**FUTURE-WORK APPENDIX**




- 14 planar controls; largest checked gap error 4.28 nm
- 10 endpoint sequences; two injected faults abort
- 84 full-solid pose checks; residuals bounded at 1 µm
- Final COM refinement changes 1.51 / 1.47 mm; no precision rating

Calibrated planar controls and sampled geometry do not establish continuous loaded clearance or hardware reliability.